

Functional and non-functional requirements

Requirements	Functional	Non-functional
The system shall display a home overview dashboard with market summary information.	X	
The dashboard shall load quickly and support customizable user views.		X
The markets dashboard shall show live stock prices, indices, currency rates and market trends.	X	
Market data shall update in real time with low latency and high accuracy.		X
The order book shall display buy/sell orders, market depth and trade information.	X	
The system shall handle large transaction volumes with high performance.		X
The news and macro module shall provide financial news, economic indicators and macro events.	X	
The system shall use reliable data sources and provide quick updates.		X
The global tracking map shall track aircraft, vessels and global movement data.	X	
Map rendering shall be scalable and provide real-time tracking.		X
The risk and analytics module shall perform risk analysis, stress testing and Monte Carlo simulations.	X	
Risk calculations shall be accurate and completed with high computation speed.		X
The compliance module shall monitor rules, transactions and regulatory requirements.	X	
The system shall provide secure access, audit readiness and data privacy.		X
The options chain shall display option contracts, prices, Greeks and expiry details.	X	
Options calculations shall be performed in real time with high accuracy.		X
The portfolio management module shall track investments, returns and holdings.	X	

Portfolio data shall remain consistent and transactions shall be secure.		X
The backtester shall test investment strategies using historical data.		X
Backtesting shall provide accurate results with high processing capability.		X
The sentiment dashboard shall analyze market sentiment from news and social data.		X
The system shall support scalable NLP processing		X
The bond screener shall filter bonds based on user criteria.	X	
The system shall provide efficient search and handle large databases.		X
The tax lot tracker shall manage investment tax records.		X
The system shall maintain data security and compliance requirements.		X

Part 3: User requirements and system requirements

We are developing a financial markets platform for a trading and investment firm that provides clients with real-time market data, analytics and portfolio tools to manage their trading activities online. This platform requires to work as follows:

The platform provides a home overview dashboard summarizing market activity, along with a markets dashboard that shows live stock prices, indices, currency rates and market trends. An order book displays buy/sell orders, market depth and trade information, while a news and macro module delivers financial news, economic indicators and macro events. A global tracking map shows aircraft, vessels and other global movement data relevant to market activity. A risk and analytics module perform risk analysis, stress testing and Monte Carlo simulations, and a compliance module monitors trading rules, transactions and regulatory requirements. Clients can view an options chain showing option contracts, prices, Greeks and expiry details, and a portfolio management module that tracks their investments, returns and holdings. A back tester allows clients to test investment strategies using historical data, and a sentiment dashboard analyses market sentiment drawn from news and social data.

The platform needs to support real-time data updates with low latency, handle large transaction volumes with high performance, and provide secure access, audit readiness and data privacy for all users. It also needs to deliver accurate, high-speed risk and options calculations, scalable NLP processing for sentiment analysis, and efficient search across large databases, while maintaining data security and regulatory compliance throughout.

- Who are the stakeholders of the system described above?
 1. Clients
 2. Developers
 3. Client Supports

- What are the non-functional requirements of the system described above?
 1. The dashboard shall load quickly and support customizable user views.
 2. Market data shall update in real time with low latency and high accuracy.
 3. The system shall handle large transaction volumes with high performance.
 4. The system shall use reliable data sources and provide quick updates.
 5. The website shall be developed using HTML, CSS and Python

 6. Risk calculations shall be accurate and completed with high computation speed.
 7. The system shall provide secure access, audit readiness and data privacy.
 8. Options calculations shall be performed in real time with high accuracy.
 9. Portfolio data shall remain consistent and transactions shall be secure.
 10. Back testing shall provide accurate results with high processing capability.

- Write down the user and associated system requirements for the system described above in natural language.
 - The system shall allow clients to view live market data including stock prices, indices and currency rates.
 - The system shall display market news and macroeconomic indicators to all users.
 - The system shall display order book details for every traded instrument.
 - The system shall generate personalized portfolio insights and investment recommendations based on a client's trading history.
 - The system shall allow the clients to browse available instruments and market data.